SwiftLog Machine Learning: Executive Summary

Week 5 | Machine Learning Track | Gyimah Ramsey Opoku

The question

Can machine learning predict late deliveries, product returns, demand and customer value from SwiftLog's operational data, so that management can act earlier?

The answer

Not with the data collected today. Four models were built and tested properly (five algorithms each, tuned, cross-validated, compared to baselines). None performs better than a naive guess. Delay and return predictions score a ROC-AUC of 0.52 and 0.49, where 0.50 is a coin flip, and the demand and customer value forecasts explain none of the variation (R-squared -0.06 and -0.003). Independent statistical tests, and models trained on scrambled data, reach the same conclusion: the fields available carry no usable signal.

Model	Result on unseen data	vs. baseline
Delivery delay	AUC 0.52 (95% range 0.48 to 0.56)	Same as chance
Product return	AUC 0.49 (95% range 0.42 to 0.56)	Same as chance
Demand forecast	R-squared -0.06; error 11.3 units/product-month	Same as predicting the average
Customer value	R-squared -0.003	Same as predicting the average

What management should take from this

- **Do not act on the current predictions.** They would direct staff and stock using noise.
- **The delay problem is real and company-wide:** 63.7% of shipments are 3+ days late, with no warehouse standing out. This calls for a process review.
- **Stock is far above demand:** about 150 months of cover for the median product. Check the data, then release working capital.
- **63% of returns are wrong-item or damaged goods:** picking and packing quality, not product choice, is the lever.
- **Fix the data:** add timestamps, carrier, route, stock movements and a supplier link. Supplier reliability could not be modelled at all without it.

What was delivered

A leakage-safe, re-runnable ML pipeline; a catch of two data problems that would have produced false results (a duplicated delay column and a missing supplier link); explainability analysis; a demo web application; a full technical report and presentation; and twelve prioritised recommendations. When richer data arrives, the same workflow can be re-run unchanged to test again.

Next steps

- Month 1: root-cause review of delays and returns; start capturing the missing fields.

- Months 2 to 4: pilot an alternative dispatch rule on a random share of orders to create evidence.
- After 6 to 12 months of richer data: re-run the notebook and deploy only if a model beats the baseline.

SwiftLog Logistics Limited is a fictional client; all data is synthetic and created for the AnalystLab Africa Experience Lab.